Partha Chakraborty

(716) 468 9420 | parthac350@gmail.com | Portfolio | LinkedIn | GitHub

SUMMARY

Software engineer with 2+ years at startups building API integrations, workflow automation, and LLM features on Python, FastAPI, Node.js, and React, plus a deployed multi agent RAG app. M.S. in AI and Machine Learning, University at Buffalo, expected January 2027.

EXPERIENCE

TurboMail **Remote**
AI Engineer Intern *May 2026 to Aug 2026*

- Wrote bulk email deletion and folder move APIs in FastAPI, plus email body sanitization and multi account thread scoping.
- Automated unsubscribe handling with FastAPI background tasks that parse RFC 8058 List Unsubscribe headers.
- Designed the database schema and APIs for subscription and payment tracking, then built the customer facing React interface.
- Owned production features across the stack in React, TypeScript, Python, FastAPI, SQLAlchemy, and SQL.
- Improved accuracy by 70% relative to the base Qwen2.5 7B model by fine tuning with LoRA and PEFT in PyTorch on a custom email dataset for an in product LLM feature.

Bird Vision AB **Remote**
Software Developer *Jun 2023 to Jul 2025*

- Integrated Eboka with the company's other products end to end, so customers got value from each in one subscription cycle.
- Built two way customer data sync between two company apps, with scheduled background bulk imports that notify users.
- Added on demand single customer import by user ID in both directions, so users didn't have to enter the same data twice.
- Took backend features to production, connecting third party APIs to existing workflows with Node.js and Express.js.
- Ranked second on the team in code contributions within a year, shipping frontend and backend work across three products.
- Wrote Cypress tests covering the main user flows in the web app.

EDUCATION

University at Buffalo **New York, USA**
M.S. in Artificial Intelligence and Machine Learning *GPA 3.9/4.0 | Expected Jan 2027*
Coursework: Algorithms, Database Systems, Data Intensive Computing, Machine Learning

Presidency University **Bengaluru, India**
B.Tech. in Computer Science and Engineering *GPA 8.14/10 | 2018 to 2022*
Coursework: Computer Networks, Data Structures, Operating Systems, Computer Architecture, Engineering Mathematics

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java
Backend and integrations: FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, Redis, API integration
AI and LLMs: Claude API, Gemini API, agent orchestration, agent evaluation, RAG, tool calling, Sentence Transformers, embeddings, vector search, BM25, cross encoder reranking, PyTorch, LoRA, PEFT
Data and tools: PostgreSQL, MySQL, MongoDB, Firebase, ChromaDB, Docker, Linux, Git, GitHub Actions, Pandas, Streamlit
Frontend and testing: HTML, CSS, React, Next.js, Redux, Tailwind CSS, React Native, Cypress, pytest
Development tools: Tree sitter, Cursor, Claude Code, GitHub Copilot

PROJECTS

AI Codebase Intelligence [Live Demo] [GitHub] Python, Gemini, ChromaDB, Streamlit

- Deployed a multi agent RAG app on the Gemini API that answers questions about public GitHub and local repositories through four agents: Architect, Code Reviewer, Bug Finder, and Documentation.
- Grounded answers in source code with Tree sitter chunking across six languages, hybrid semantic, BM25, and symbol retrieval, and cross encoder reranking.
- To keep answers reliable, built a source explorer with file paths, line ranges, and citation checks; reached 87.5% Hit@5 and 0.750 MRR on an initial 8 question retrieval evaluation.

Greenlight [Live Demo] [GitHub] Python, Claude, Gemini, Streamlit, GitHub Actions

- Built a Claude/Gemini agent with tool calling to diagnose failing tests, edit code, and independently verify fixes.
- Restricted tools to the repository, blocked secret files, scrubbed test process credentials, and kept tests read only by default; enforced step, token, and cost budgets with per run traces and reviewable diffs.
- Shipped a CLI, Streamlit demo, and GitHub Action for human reviewed fix PRs; developed the project with Claude Code.
- Built hidden test and regression checking evaluations: a Gemini run solved 10/10 synthetic tasks and 4/13 attempted real bugs, requiring passing project tests and no test file edits.

CERTIFICATIONS

Hugging Face Agents Course, Unit 1: Fundamentals of Agents [Certificate] Oct 2026